

# VEGANISM DEBATE MASTER SHEET

Nutrition • Ethics • Animal Use • Sentience • Ecology • Resources • Public Health • Fishing • Economics • Logic

## 1. NUTRITION & HUMAN HEALTH

- **Adequacy:** Appropriately planned vegan diets can be nutritionally adequate for adults and may support long-term cardiometabolic health. [1]
- **Protein:** Plant foods provide all essential amino acids; vegan diets can meet protein needs. Protein combining at every meal is unnecessary. [2]
- **Fiber/cholesterol:** Fiber is found in plant foods; meat, eggs and milk contain no fiber. Whole plant foods are essentially cholesterol-free. [3]
- **Cancer:** WHO/IARC classifies processed meat as Group 1 carcinogenic to humans and red meat as Group 2A probably carcinogenic. Group 1 reflects evidence strength, not equal risk with tobacco. [4]
- **RCT:** An 8-week randomized identical-twin trial found the vegan group had lower LDL cholesterol, body weight and fasting insulin. [5]
- **Debate caution:** Health associations do not automatically prove causation. A vegan diet still requires appropriate planning, including reliable vitamin B12.

## 2. ETHICS & MORAL PHILOSOPHY

- **Core question:** If a sentient being can suffer, and using or killing that being is not necessary, what morally relevant reason justifies the harm?
- **Key distinctions:** Natural ≠ ethical. Legal ≠ ethical. Tradition ≠ justification. Preference ≠ necessity. Explanation ≠ justification.
- **Rights vs welfare:** Welfare asks how animals are treated; animal rights asks whether humans are justified in using animals as resources at all.
- **Consistency:** Species membership alone must be defended as a morally relevant distinction if it is used to discount comparable interests in avoiding pain and suffering.

## 3. SENTIENCE & COGNITION

- **Fish:** Evidence reviews report pain, fear/anxiety, learning and memory in fish. [6]
- **Other animals:** Research describes learning, memory, social cognition and/or emotions in pigs, chickens and cattle. [7]
- **Aquatic invertebrates:** Scientific reviews support sentience evidence in cephalopods and decapod crustaceans. [8]
- **Debate point:** Sentience ≠ intelligence. The capacity for subjective experience is distinct from advanced reasoning.

## 4. ANIMAL USE, FARMING & SLAUGHTER

- **Piglets:** USDA sources document painful procedures including castration, tail docking, teeth clipping and ear notching/tagging; USDA research reports these are usually performed without pain mitigation. [9]
- **Castration/dehorning:** USDA materials describe castration as painful and document pain/distress following dehorning. [10]
- **Dairy:** Commercial milk production depends on cows becoming pregnant and giving birth; dairy calves are commonly separated from their mothers soon after birth. [11]
- **Chickens:** USDA guidance describes live birds hung upside down in shackles, electrically stunned, neck vessels cut and bled. Poultry are not covered by the federal Humane Methods of Slaughter Act. [12]
- **Transport/confinement:** EFSA identifies welfare hazards including heat stress, thirst, fatigue, pain, fear, distress and restrictions on highly motivated natural behaviors. [13]

## 5. ECOLOGY & ENVIRONMENT

- **Land:** Meat and dairy use ~83% of farmland while providing ~18% of calories and ~37% of protein; a no-animal-product scenario used ~76% less agricultural land. [14]
- **Climate:** Food systems produce roughly one-third of human-caused GHG emissions; one global analysis attributed ~57% of food-production GHGs to animal-based foods. [15]
- **Diet shift:** Poore & Nemecek estimated eliminating animal products could reduce food-related GHG emissions ~49%, acidification ~50% and eutrophication ~49%. [14]
- **Deforestation:** FAO estimates agriculture drives ~90% of global deforestation; cropland expansion and livestock grazing are major components. [16]
- **Water/pollution:** Agriculture accounts for ~72% of global freshwater withdrawals. Manure contributes nitrogen/phosphorus runoff, while livestock ammonia contributes to harmful PM2.5. [17]

## 6. FOOD & RESOURCE EFFICIENCY

- **Feed conversion:** Using human-edible crops as animal feed loses calories compared with direct human consumption. [18]
- **Crops:** Large shares of global soy and cereals are used for animal feed, so livestock production can add crop demand rather than eliminate it. [19]
- **Crop-deaths objection:** Crop production can harm animals, but feeding crops through animals adds a trophic step and generally increases land/feed requirements; the relevant comparison is total harm and resource use, not whether any food system is harm-free.

## 7. PUBLIC HEALTH

- **Antimicrobial resistance:** Antibiotic misuse/overuse accelerates resistance; antimicrobial use in livestock can select for resistant microbes that move among animals, humans and ecosystems. [20]
- **Zoonoses:** A major global analysis found most emerging infectious-disease events in its dataset originated in animals; livestock intensification can affect zoonotic-disease risk. [21]

## 8. FISHING & AQUACULTURE

- **Bycatch:** Commercial fishing unintentionally catches, injures and kills non-target fish, sea turtles, marine mammals and seabirds. [22]
- **Aquaculture:** Aquaculture feeds can use wild fish, fishmeal and fish oil; fisheries supplying feed can themselves produce bycatch. [23]
- **Ethical relevance:** Fish sentience evidence means fishing debates concern animal welfare as well as conservation and sustainability. [6]

## 9. ECONOMICS

- **Diet cost:** Modeling found plant-based diets can cost less in high-income countries, with vegan diets among the cheapest modeled healthy diets. [24]
- **Societal effects:** Global vegan scenarios have modeled large health and climate-related economic savings by 2050; these are scenario estimates, not observed savings. [25]

## 10. COMMON ARGUMENTS / LOGIC CHECK

- **"Plants feel pain."** A response to stimuli is not by itself evidence of subjective pain. Also ask whether feeding crops to animals reduces or increases total crop use.
- **"We are omnivores."** Ability to eat animal foods does not establish nutritional necessity or ethical justification.
- **"It is natural/traditional."** Describing what occurs in nature or history does not establish what ought to be done.
- **"Veganism isn't perfect."** A solution need not eliminate every harm to reduce avoidable harm; perfection is a different standard from improvement.
- **"It's personal choice."** When a choice affects another sentient being, the debate concerns the interests of everyone affected, not only the chooser.
- **"Animals eat animals."** Nonhuman animal behavior does not itself establish human moral obligations or justify choices humans can evaluate.
- **Debate discipline:** Assertion ≠ evidence. Anecdote ≠ scientific evidence. Correlation ≠ causation. Possibility ≠ probability. Disagreement ≠ refutation.

## SOURCE KEY

[1] Academy of Nutrition & Dietetics 2025, doi.org/10.1016/j.jand.2025.02.002 [2] Mariotti & Gardner 2019, doi.org/10.3390/nu11112661 [3] NIH MedlinePlus / USDA FoodData Central [4] WHO/IARC processed & red meat Q&A [5] Landry et al. 2023, doi.org/10.1001/jamanetworkopen.2023.44457 [6] Lambert et al. 2022, doi.org/10.3390/ani12091182; Brown 2015 [7] Marino/Colvin; Marino; Marino/Allen [8] Birch et al., LSE 2021 [9-12] USDA APHIS/ARS/AMS/NAL sources in evidence library [13] EFSA Animal Welfare [14] Poore & Nemecek 2018, doi.org/10.1126/science.aag0216 [15] Crippa et al. 2021; Xu et al. 2021 [16] FAO Global Remote Sensing Survey [17] FAO; EPA; Domingo et al. 2021 [18] Cassidy et al. 2013 [19] FAO/FCRN/USDA data [20] WHO; WOA [21] Jones et al. 2008; Gilbert et al. 2021 [22] NOAA Fisheries [23] Global aquaculture-feed review, PMC11482329 [24] Springmann et al., Lancet Planetary Health 2021 [25] Springmann et al., PNAS 2016. Full URLs and claim-specific citations are in the 12-page Scientific Evidence Library.

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